

Pavan Kumar Rao

Senior Gameplay Programmer

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PROFESSIONAL SUMMARY

Senior Gameplay Programmer, 8+ years shipping gameplay systems in C++, Unreal Engine 5, and Unity3D. Strong across the Gameplay Ability System (GAS), character movement, AI, and networked multiplayer.

PROFESSIONAL EXPERIENCE

Senior Gameplay Programmer

Dream Game Studios | December 2022 – Present

- Architected core **character control** for Dream Cricket (batting, fielding, networked movement) using **client-side prediction** and **server reconciliation**.
- Delivered **5 major gameplay features** end to end (Async Multiplayer, Gacha, Team Management, Inventory, Emotes), from design handoff to release.
- Designed a **unified game mode architecture** powering PvP and PvE across Quick Match, Tournaments, and Private Lobby.
- Built **real-time parameter tuning tools** and gameplay debug overlays that sped up design iteration.
- Introduced an **MVCS frontend architecture** that improved codebase maintainability.
- Mentored junior engineers on code quality and established **CI/CD automation via Jenkins**.

Game Developer (SE 2)

Moonfrog Labs | January 2020 – November 2022

- Led the migration of live multiplayer title Ludo Club from **Cocos2d-x to Unity3D**, refactoring the networked game loop around a **finite state machine**.
- Shipped **full-stack multiplayer features** across client (**C#**) and server (**Node.js**), including synchronised achievement and rewards systems.
- Built **developer and admin tools** that sped up debugging of networked gameplay issues in live ops.
- Maintained live production stability through **real-time monitoring**, managed deployments, and rapid bug resolution.

Associate Game Developer

JumpStart Games | January 2018 – December 2019

- Contributed gameplay and social features across multiple shipped **Unity3D** titles, including *Neopets: Legends and Letters*.
- Worked on synchronised **game state** and social systems, and supported live ops with bug fixes and feature updates.

KEY PROJECTS

Unreal First Person Hack-and-Slash (C++ / Unreal Engine 5)

- Built a **GAS-driven combat system** with ability management, cooldowns, and attribute replication, driving animation montages and combat-state blending.
- Built reactive enemy AI with custom **State Trees** for decision-making, combat behaviour, and group tactics.
- Combined **UE5 Blueprints** with C++ for rapid prototyping of gameplay and animation logic.

Top-Down Multiplayer Shooter (Unity3D)

- Implemented **client-side prediction**, **server reconciliation**, and lag compensation over peer-to-peer and client-server architectures.
- Synchronised networked movement and combat with **entity interpolation** for smooth animation playback.

Unity AI Module (C# / Unity)

- Built a custom **A* pathfinder** optimised with a binary-heap priority queue for large environments.
- Developed a **behaviour tree framework** with a custom graph editor for visual debugging and tuning.

2D C++ ECS-Based Engine (C++)

- Designed a custom **ECS framework** with multithreading and a job system for parallel entity processing.
- Built a **sequential-impulse physics system** for collision resolution and rigid-body dynamics.
- Applied **linear algebra**, 3D math, and manual memory management throughout the engine.

TECHNICAL SKILLS

Engines	Unreal Engine 5 (C++ & Blueprints), Unity3D (C#)
Core Systems	GAS, Character Movement, Enhanced Input, Camera Systems, AI State Trees, Behaviour Trees
Animation	Animation Montages, State Machines, Animation Blueprints, Networked Animation Sync, Entity Interpolation
Multiplayer	Client-server architecture, client-side prediction, server reconciliation, lag compensation
Languages	C++, C#, JavaScript (Node.js), Python, GLSL
Math & Low-Level	Linear Algebra, 3D Math, ECS, Multithreading, Job Systems, Physics Systems
Tools	Git, Perforce, Jenkins (CI/CD), Agile/Scrum

EDUCATION

B.Sc (Hons) in Game Development

Bharathiar University | 2014 – 2017